Solutions

Question 7.1: Assuming that there is no friction acting against the object, the component of force parallel to the plane ($F2$) is the only force acting on the object, which is $mg\sin(30)$. As we want to estimate the velocity after it has moved $10m$ down the incline, we use ODE (ii) from above:

$$mv\frac{dv}{dx} = mg\sin(30)$$

Cancel the mass (m) and use separation of variables to integrate. In terms of the limits, we want to see estimate the velocity after it has rolled $10m$ down the incline from rest so we integrate x between 0 and 10, and integrate v from 0 to v:

$$\begin{aligned}
\int_0^v v\,dv &= g\sin(30)\int_0^{10} dx \\
\left[\frac{1}{2}v^2\right]_0^v &= g\sin(30)\times[x]_0^{10} \\
\frac{1}{2}v^2 &= 10*g\sin(30) \\
v^2 &= \sqrt{20*g\sin(30)} = 9.9
\end{aligned}$$

So the object was moving at $9.9m/s$ after it had moved a distance of 10 metres.

Question 7.2: This one is very similar to the above question, but now we have some force of friction. The force parallel to the plane ($F2$) is acting on the object and friction of 0.1 times the normal force ($F1$) so the total forces acting on the object is:

$$\begin{aligned}
F &= F2 - 0.1F1 \\
&= mg\sin(30) - 0.1mg\cos(30)
\end{aligned}$$

Again, as we want to estimate the velocity after it has moved $10m$ down the incline, we use ODE (ii) from above:

$$mv\frac{dv}{dx} = mg\sin(30)$$

Cancel the mass (m) and use separation of variables to integrate x between 0 and 10, and integrate v from 0 to v (since it was at rest at $x = 0$):

$$\begin{aligned}
\int_0^v v\,dv &= (g\sin(30) - 0.1g\cos(30))\int_0^{10} dx \\
\left[\frac{1}{2}v^2\right]_0^v &= (g\sin(30) - 0.1g\cos(30))\times[x]_0^{10} \\
\frac{1}{2}v^2 &= 10(g\sin(30) - 0.1g\cos(30)) \\
v^2 &= \sqrt{20(g\sin(30) - 0.1g\cos(30))} = 9.0
\end{aligned}$$

So the object was moving at $9.0m/s$ after it had moved a distance of 10 metres.

Question 7.3: This one is very similar to the above question, but we need to calculate velocity in terms of time instead of distance. Like the last question, the force parallel to the plane ($F2$) is acting on the object and friction of 0.1 times the normal force ($F1$) so the total forces acting on the object is:

$$\begin{aligned} F &= F2 - 0.1F1 \\ &= mg\sin(30) - 0.1mg\cos(30) \end{aligned}$$

As we want to estimate the velocity 10 seconds after it moved down the incline, we use ODE (i) from above:

$$m\frac{dv}{dt} = mg\sin(30)$$

Cancel the mass (m) and use separation of variables to integrate. In terms of the limits, we want to estimate the velocity 10 seconds after it has moved down the incline from rest so we integrate t between 0 and 10, and integrate v from 0 to v:

$$\begin{aligned} \int_0^v dv &= (g\sin(30) - 0.1g\cos(30))\int_0^{10} dt \\ [v]_0^v &= (g\sin(30) - 0.1g\cos(30)) \times [t]_0^{10} \\ v &= 10(g\sin(30) - 0.1g\cos(30)) = 40.5129 \end{aligned}$$

So the object was moving at $40.51m/s$ after 10 seconds.

Question 7.4: This Question 7.is similar to Question 7.1 but instead of starting from rest the object is moving at $2m/s$. As we want to estimate the velocity after it has rolled $10m$ down the incline, we use ODE (ii) from above:

$$mv\frac{dv}{dx} = mg\sin(30)$$

Cancel the mass (m) and use separation of variables to integrate. In terms of the limits, we want to estimate the velocity of the object which had an initial velocity of $2m/s$ after it has rolled $10m$ down the incline from rest so we integrate x between 0 and 10, and integrate v from 2 to v:

$$\begin{aligned} \int_2^v v\,dv &= g\sin(30)\int_0^{10} dx \\ \left[\frac{1}{2}v^2\right]_2^v &= g\sin(30) \times [x]_0^{10} \\ \frac{1}{2}v^2 - \frac{1}{2}(2)^2 &= 10 * g\sin(30) \\ \frac{1}{2}v^2 - 2 &= 10 * g\sin(30) \\ v^2 &= \sqrt{20 * g\sin(30) + 4} = 10.1 \end{aligned}$$

So the object with an initial velocity of $2m/s$ was travelling at $10.1m/s$ after it had rolled down a distance of 10 metres.

Question 7.5: Since the sphere is moving horizontally, there is no force acting on the object in the direction of travel. As the wind speed is zero, the total forces on the object is:

$$\frac{\rho v^2}{2}\pi r^2$$

We want to estimate the distance the sphere with an initial velocity of $10m/s$ travels before slowing down to $0.01m/s$, we use ODE (ii) from above:

$$mv\frac{dv}{dx} = -\frac{\rho v^2}{2}\pi r^2$$

Using separation of variables we can integrate this ODE. In terms of the limits, we want to estimate the distance travelled as the sphere goes from $10m/s$ and $0.01m/s$, so we integrate x between 0 and x, and integrate v from 10 to 0.01:

$$\begin{aligned}
\frac{m}{v}\,dv &= -\frac{\rho}{2}\pi r^2\,dx \\
\int_{10}^{0.01}\frac{m}{v}dv &= -\int_0^x \frac{\rho}{2}\pi r^2 dx \\
m\left[\ln v\right]_{10}^{0.01} &= -\frac{\rho}{2}\pi r^2\left[x\right]_0^x \\
m\left[\ln 10 - \ln 0.01\right] &= \frac{\rho}{2}\pi r^2 x \\
x &= \frac{2\times 6.91m}{\rho\pi r^2} \cong 14.66\text{m}
\end{aligned}$$

So the sphere with an initial velocity of $10m/s$ had moved $14.66m$ when it reached a velocity of $0.01m/s$.

Question 7.6: We can relate an object moving with initial velocity v_0 and end velocity v_1 and a starting position $x = 0$ through:

$$mv\frac{dv}{dx} = ma$$

Since the object is moving with constant acceleration then we can just use separation of variables and integrate. In terms of the limits we integrate x between 0 and x and integrate

v between v_0 and v_1:

$$\begin{aligned} v\frac{dv}{dx} &= a \\ \int_{v_0}^{v_1} v\,dv &= \int_0^x a dx \\ \left[\frac{1}{2}v^2\right]_{v_0}^{v_1} &= a\,[x]_0^x \\ \frac{1}{2}(v_1^2 - v_0^2) &= 2ax \\ v_1^2 - v_0^2 &= 2ax \end{aligned}$$

Question 7.7: We can relate an object moving with initial velocity v_0 at time $t = 0$ through:

$$m\frac{dv}{dt} = ma$$

Since the object is moving with constant acceleration then we can just use separation of variables and integrate. In terms of the limits we integrate t between 0 and t and integrate v between v_0 and v_1:

$$\begin{aligned} \frac{dv}{dt} &= a \\ \int_{v_0}^{v} dv &= \int_0^t a dt \\ v - v_0 &= at \end{aligned}$$

Question 7.8: We need to find the highest point a ball thrown into the air reaches. The ball has an initial velocity of $10m/s$ and will reach its highest point where it has a velocity of $0m/s$. Since the object has constant acceleration we can use the general formula we developed in Question 7.7 with $v_0 = 10$, $v_1 = 0$, and $a = -9.8$ we see that

$$\begin{aligned} 2ax &= v_1^2 - v_0^2 \\ x &= \frac{v_1^2 - v_0^2}{2a} \\ x &= \frac{0^2 - 10^2}{2 * (-9.8)} \\ x &= \frac{-100}{-19.6} = 5.101 \end{aligned}$$

so the ball reached $5.1m$ high.

Question 7.9: This is similar to the last question, but we need to find the time it takes to find its maximum height. The ball has an initial velocity of $10m/s$ and will reach its highest point where it has a velocity of $0m/s$. Since the object has constant acceleration

we can use the general formula we developed in Question 7.8 with $v_0 = 10$, $v_1 = 0$, and $a = -9.8$ we see that

$$\begin{aligned} at &= v - v_0 \\ t &= \frac{v - v_0}{a} \\ t &= \frac{0 - 10}{-9.8} \\ t &= \frac{-10}{-9.8} = 1.02 \end{aligned}$$

so the ball reached its maximum height after $1.02s$.